

//make sure that all identifiers meaning constants, lists/arrays and variables using camel case

Age = 14

Age ← 14

First_Name ← "Name"

Last_Name ← "Last Name"

Names_Collected ← INPUT "enter your last name"

Last_Name ← Names_Collected

Equal to - = //in python it would be ==, one = sign in python be the equivalent of ← assigning a piece of data

An algorithm is merely the sequence of steps taken to solve a problem.

Operator	Meaning/purpose	Example	Result
=	<u>Equal</u> – tests if two expressions are identical	IF a=8 IF name\$="JANET" IF a=14	True False False
<	<u>Less Than</u> – tests if the first expression is less than the second	IF a<8 IF name\$<"JANET" IF a<14	False True True
>	<u>Greater Than</u> – tests if the first expression is greater than the first	IF a>8 IF name\$>"JANET" IF a>14	False False False
<>	<u>Not Equal</u> – tests if two expressions are different	IF a<>8 IF name\$<>"JANET" IF name\$<>"JAMES" IF a<>14	False True False True
<=	<u>Less Than or Equal</u>	IF a<=8 IF name\$<="JANET" IF a<=14	True True True
>=	<u>Greater Than or Equal</u>	IF a>=8 IF name\$>="JANET" IF a>=14	True False False

Write a program that compares two different numbers. If number 1 is bigger than number two, we want the program to output "number 1 is bigger", otherwise we want them to enter the number again until number 1 is bigger than number 2.

Num1 ← INPUT "Enter a number"//this is how to to setup a variable.

Num2 ← INPUT "Enter a number"

IF Num1 > Number2

 OUTPUT "number 1 is bigger"

ELSE

 OUTPUT "number 1 is not bigger than number 2, try again"

 Num1 ← INPUT "Enter a number"

Name ← INPUT "Enter a programmer name"

IF Name = "Essa" OR "essa" OR "ESSA"

 OUTPUT "great programmer"

IF Name = "Rob" or ELSE

 OUTPUT "terrible programmer"

```
IF { conditional statement } THEN
    { statement 1 }
ELSE
    { statement 2 }
ENDIF
```

IF {conditional statement} THEN

 OUTPUT {statement 1}

ELSE

 OUTPUT {statement 2}

ENDIF

Loops

,

FOR count ← 1 TO 5000

 INPUT House "Enter your house value"

 IF House > 50 000 THEN

 Tax← House * 0.010

 ELSE IF House > 100 000 THEN

 Tax ← House * 0.015

 ELSE IF House > 200 000 THEN

 Tax← House * 0.020

 ELSE

 Tax ← 0

 OUTPUT Tax

NEXT

```

X ← TRUE
WHILE X ← TRUE
    INPUT House "Enter your house value"
    IF House > 50 000 THEN
        Tax ← House * 0.010
    ELSE IF House > 100 000 THEN
        Tax ← House * 0.015
    ELSE IF House > 200 000 THEN
        Tax ← House * 0.020
    ELSE
        Tax ← 0
    OUTPUT Tax
WEND

```

```

INPUT X "Enter your value: "

```

```

WHILE X != 0 THEN
    N ← X * X / (1 - X)
    X -= 1
    OUTPUT N

```

To set up an array in pseudocode, you do the following:

```

Identifier ← [ ] // identifier is name of the array

```

An alternative method is:

```

DECLARE Identifier ← [ ]

```

A 1D array would look like this

```

DECLARE Names ← [ ]

```

If you know the size of the array it would be:

```

DECLARE Names ← [1:30]

```

A 2D array would look like this:

```
DECLARE Names ← [[1:30],[1:30]]//arrays begin at index 1 in pseudocode
```

An array with existing elements would like this:

```
Names ← ["Jeet", "Jord", "Jeremy"]
```

To add a new name to the array, you would use one of the following:

```
Name ← USERINPUT "Enter your name"
```

```
Names ← Names + Name //add the variable after the array name
```

Another method you could use is:

```
Names.APPEND(Name)
```